Reply Mirror

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Abstract

In the futuristic digital metropolis of Reply Mirror, the financial institutions MirrorPay are synonymous of reliability and speed: cloud data platforms capable of processing billions of transactions for millions of customers.

In this highly interconnected world, most transactions are monitored, data is widely accessible and privacy is increasingly limited. As key managers of economic activity, you guide financial flows and support trust in digital systems.

As a member of The Eye, you are enrolled to intelligently, responsively and dynamically monitor the system. So, let's build an autonomous intelligence able to observe, adapt and act when the system comes under siege.

The mission is clear: when artificial intelligence becomes the weapon of fraudsters, only an even more advanced intelligence can restore balance.

1 Problem Statement

It is 2087 and the digital metropolis of Reply Mirror thrives on transparency: almost every piece of personal data is publicly accessible. Financial institutions like MirrorPay not only manage the flow of economic transactions, but also hold vast troves of demographic and various data about their citizens. Yet in this data-rich world, new dangers lurk beneath the surface: fraudulent behaviour hides in plain sight, blending seamlessly with legitimate activity.

Among this growing complexity, you — The Eye — take the stage. Your mission is vital: to detect and neutralize financial fraud, preserving the integrity of the system and protecting honest citizens.

The malicious actors — the so-called Mirror Hackers — are smart, adaptive and relentlessly strategic. They never follow a fixed pattern; instead, they constantly reshape their pattern to outsmart detection systems. Their evolving tactics include:

- targeting new merchants and transaction categories;
- shifting temporal habits (e.g. moving from daytime to late-night activity);
- operating across changing geographic regions and jurisdictions;
- varying transaction amounts and frequency;
- creating new, deceptive behavioural sequences.

At the heart of the challenge lies the constant evolution of fraudulent activity over time. Success depends on your system's ability to learn, adapt and generalize as the landscape changes. Static models will fail; only dynamic, continuously learning and strategically adaptive solutions can stay ahead in this ever-shifting battlefield.

Challenge Flow The challenge unfolds in five levels. At each level, your team receives a training dataset representing the economic transactions between citizens of ReplyMirror. In addition to this, you are provided with additional datasets containing varying levels of information about citizens, their communications and their habits.

In addition to the training data, each level also includes a corresponding evaluation dataset of comparable difficulty. These evaluation datasets are the only ones used to compute the official score for the level. For each evaluation dataset, only the first submission will be accepted and will be considered final.

Throughout all levels, you are tasked with designing a system of cooperative intelligent agents capable of identifying anomalous activities in environments where fraudulent strategies evolve over time. Using the available data, your team must determine for every transaction in the dataset whether it is legitimate or fraudulent.

The score for each round depends on the system's ability to adapt to new conditions while maintaining consistent accuracy. At the end of all five levels, the overall leader-board will reflect the resilience and adaptability of the system of each team.

Challenge Goal You are asked to design an agent-based system capable of:

- detecting fraudulent behaviour that evolves over time and blends seamlessly with legitimate transactions;
- anticipating new attack patterns by leveraging the memory of past interactions;
- responding in real time to sudden changes without degrading performance;
- keeping the false positive rate low, avoiding the unnecessary blocking of legitimate activity.

Each incorrect decision carries an asymmetric cost:

- blocking a legitimate transaction (false positive) incurs economic and reputational losses;
- allowing or ignoring a fraudulent transaction (false negative) causes significant financial damage.

The final score is calculated by combining accuracy, temporal stability and adaptability to structural changes in the data.

2 Input Format

The input consists of a series of data set with increasing complexity. You have access to multiple types of data set, available for download in different formats. In detail:

- ***Transactions.csv*** with T records of financial transactions, including:
  - Transaction ID: a unique identifier of the transaction;
  - Sender ID: a unique identifier of the transaction sender;
  - Recipient ID: a unique identifier of the transaction recipient;
  - Transaction Type:
    * bank transfer
    * in-person payment
    * e-commerce
    * direct debit
    * withdrawal
  - Amount of the occurred transaction;
  - Location (provided only for in-person payments);
  - Payment Method:
    * debit card
    * mobile device
    * smartwatch
    * GooglePay
    * PayPal
  - Sender IBAN (provided only for bank transfers);
  - Recipient IBAN (provided only for bank transfers);
  - Balance: remaining amount after the transaction;
  - Timestamp of the occurred transaction;

- ***Locations*** with geo-referenced data, whereby the citizen has been located through GPS-based systems, irrespective of the type of activity performed, including:
  - BioTag: a unique identifier of the citizen;
  - Datetime: date and time of geo-referentiation;
  - Lat: latitude coordinate in degrees;
  - Lng: longitude coordinate in degrees;
- ***Users*** with a resume of citizen's personal data;
- ***Conversations*** with threads of occurred conversations, including:
  - User ID: a unique identifier of the citizen;
  - SMS: complete textual thread;
- ***Messages*** with e-mail interactions, including:
  - mail: complete textual thread.

3 Output Format

The output must be an ASCII text file. Each line should be separated by newline character and refers to a suspected fraudulent Transaction ID.

The format of each line is:

$$t$$

where:

- t represents the suspected fraudulent Transaction ID among the T Transaction ID received in input.

The output is considered invalid if any of the following conditions occur:

- no transactions are reported;
- all transactions are reported;
- less than 15% of the fraudulent transactions are correctly identified.

4 Scoring Rules

The evaluation framework is built around a composite scoring model that measures performance across two key dimensions, with economic impact serving as the primary driver. Your goal is not to design a static, deterministic, high-accuracy model, but to develop an AI agent-based system that is economically sustainable, operationally efficient and capable of operating in a real-world production environment.

Accuracy This component of the score evaluates the system's capability to correctly detect fraudulent behaviour while minimizing unnecessary transaction blocks. The goal is to achieve a balanced trade-off between identifying fraud and avoiding unjustified customer impact.

Additional Metrics Cost, speed and efficiency serve as complementary evaluation metrics. They reward the design of an optimized agent architecture capable of detecting fraud in real time while maintaining low operational expense.

These dimensions emphasize the system's ability to perform under real-world constraints, balancing computational resources, processing latency and infrastructure usage. In essence, they measure how effectively your AI agent-based system can scale, respond and adapt without compromising economic sustainability or responsiveness.

5 Example

Here, you can find 3 rows from Transactions.csv file. The players suspect that two of these three transactions could be fraudulent. Respective Transaction IDs are printed in output file.

Input Example

```
4a92ab00-8a27-4623-ab1d-56ac85fcd6b0,SCHV-SVRA-7BC-COR-0,,
e-commerce,56.63,,mobile device,IT16Y9430002300167070752952,
IT35O1753705526805948017123071,603.37,,2025-11-17T00:35:29.446363

8830a720-ff34-4dce-a578-e5b8006b2976,LRNT-MTTH-7BF-PAR-1,,
prelievo,150,Turin,debit card,FR46C7104822278076244862444,
IT39S3051166323954859019873188,142.58,,2025-11-17T14:33:28.068080

1c6db202-22d8-443f-86e7-fb1a8df05e84,TRBU-MRTT-7C4-MUL-0,BTSWF98176,
e-commerce,170.33,SwiftCart Marketplace,debit card,
DE20X3656132271467727362296,IT14Q8802310964869133978727249,
54000.24,,2027-01-04T00:00:00
```

Output Example

```
4a92ab00-8a27-4623-ab1d-56ac85fcd6b0
8830a720-ff34-4dce-a578-e5b8006b2976
```

6 Requirements

Please, take note of the following requirements:

- Only agent-based solutions are permitted.
- Fully deterministic approaches will be evaluated with reservation.
- Submissions must include the actual implemented code, along with clear execution instructions and a complete list of required libraries in order to ensure the reproducibility of the results.
- Top-performing teams may be subject to a re-evaluation process on new datasets not provided during the challenge.